MATH 243 — One-Page Study Priority

Based on 12 BUET papers (2011-12 to 2023-24); the 2 MATH 243 papers are weighted double. **Red = must master** · **Blue = high value** · Grey = quick marks.

#	Topic (read in this order)	How often / weight	Read from
1	Hypothesis testing — z/t tests, two-sample t, proportions, CI, P-value, Type I/II, definitions	12/12 papers · 253 marks · 35–47 per MATH 243 paper	Textbook (Walpole ch. 9–10) + ANOVA lecture intro; t & z tables
2	Correlation & regression — two regression lines, r, prediction, rank correlation, R^2, quadratic fit	12/12 · 234 marks	Stats 7
3	Probability — bag transfer, total probability, Bayes, independence, reliability	9/12 · always Q1 now	Prob L1–L5
4	Moments, skewness, kurtosis (β_1, β_2, γ_1, γ_2) + dispersion / CV comparison	12/12 · 330 marks combined	Stats 3, Stats 4
5	Graphs — stem-and-leaf, scatter, histogram/ogive, Sturges' rule	new: 15–23 marks per MATH 243 paper	Stats 5, 6
6	ANOVA — one-way (every recent paper) and two-way	last 3 papers · 18–25 marks	Stats 8 + F-table
7	Normal distribution — z areas, reverse look-up, finding μ/σ	8/12 · ~11–23 marks	Prob L12 + normal table
8	Random variables — pdf/CDF, find k, joint & marginal/conditional, discrete joint table, covariance	15–20 marks per MATH 243 paper	Prob L6, L7, L8
9	Expectation & Chebyshev — E[g(X)], Var(aX+b), Chebyshev's theorem	11–20 marks	Prob L9 + other distributions
10	Chi-square tests — goodness of fit, contingency table	10–18 marks	Stat tables + textbook
11	Poisson (+ binomial→Poisson proof), Binomial, Exponential (memoryless), hypergeometric, multinomial	Poisson 12–22; exponential 12–15	Prob L10, L11, other distributions
12	Central tendency (missing frequencies), sampling, CLT, stratified sampling, data sources	short theory, 6–10 marks	Stats 1.1, 1.2, 2

Exam pattern (MATH 243)

• **Section A (probability):** Q1 probability/Bayes · Q2 pdf/CDF & joint distributions · Q3 binomial/Poisson + Chebyshev · Q4 normal + correlation/regression.
• **Section B (statistics):** Q5 descriptive stats & graphs · Q6 moments/skewness + CLT/sampling · Q7 hypothesis tests · Q8 chi-square + ANOVA.
• You answer 3 of 4 questions in each section, so the weakest one per section can be skipped. Never skip Q1, Q4, Q7 or Q8.

Smart moves

• Questions repeat almost word for word: GPA z-test + 99% CI, "families with 5 children" chi-square, the 10-20…50-60 table (18, 20, 30, 22, 10) for skewness, mice life-span, missing frequencies with a given median.
• Solve the 2022-23 and 2023-24 MATH 243 papers fully, under time; they best match your exam.
• Get fast at long table calculations (moments, regression, ANOVA): 15–25 marks each, pure procedure.
• Your slides have no full lecture on z/t/chi-square tests or confidence intervals; cover them from the textbook.